

UNIT-II - EXPECTATION & DISCRETE DISTRIBUTIONS

Expectation or mean:

Expectation is denoted with $E(X)$ or μ .

If X is a random variable, then expectation of X is given by $E(X) = \begin{cases} \sum x f(x) & \text{if } X \text{ is discrete random variable} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } X \text{ is continuous random variable} \end{cases}$

Variance:

(Mean)

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \begin{cases} \sum x^2 f(x) - \mu^2 & \text{if } X \text{ is D.R.V} \\ \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2 & \text{if } X \text{ is C.R.V} \end{cases}$$

Q] A random variable X has the following probability function.

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Determine (i) k , (ii) $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$

(iii) If $P(X \leq k) > 1/2$ find min value of k .

(iv) Determine the distribution of x .

(v) Mean (vi) Variance.

W.K.T. $f(x) = P(X=x)$ satisfies two properties.

(i) $f(x) \geq 0$

(ii) $\sum f(x) = 1$

∴ $k \geq 0$

$$\begin{aligned} k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k &= 1 \\ 9k + 10k^2 &= 1 \end{aligned}$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$$\Rightarrow k = -1, 1/10$$

$$\text{But } k > 0$$

$$\therefore k = 1/10$$

$$(ii) P(X < 6) = P(X=0) + P(X=1) + P(X=2) + \dots + P(X=5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= \frac{8}{10} + \frac{1}{100}$$

$$= \frac{81}{100} = \underline{\underline{0.81}}$$

$$P(X \geq 6) = 1 - P(X < 6)$$

$$= 1 - 0.81$$

$$= \underline{\underline{0.19}}$$

$$P(0 < X < 5) = P(X=1) + P(X=2) + \dots + P(X=4)$$

$$= k + 2k + 2k + 3k$$

$$= 8k$$

$$= \frac{8}{10}$$

$$= \underline{\underline{0.8}}$$

$$(iii) \text{ Given that } P(X \leq k) > 1/2$$

$$P(X=0) + P(X=1) + \dots + P(X=k) > 1/2$$

$$0 + 0.1 + 0.2 + 0.2 + 0.3 + \dots > 1/2$$

$$\rightarrow 0.8 > 0.5$$

$$\therefore \underline{\underline{k=4}}$$

$$(iv) 10k^2 + 9k - 1$$

$$(v) 0 \times 0 + 1 \times k + 2 \times 2k + 3 \times 2k + 4 \times 3k + 5 \times k^2 + 6 \times 2k^2 + 7 \times k^3 + k$$

$$= k + 4k + 6k + 12k + 66k^2 + 7k$$

$$= 66k^2 + 30k$$

$$= 3.66$$

$$(vi) \sum x^2 f(x) - \mu^2$$

$$= 0 + \underline{k} + \underset{8k}{4 \cdot 2k} + \underset{18k}{9 \cdot 2k} + \underset{36k}{16 \cdot 3k} + \underset{25k^2}{25 \cdot k^2} + \underset{72k^2}{36 \cdot 2k^2} + \underset{216k^2}{49 \cdot (7k^2 + k)} - (3.66)^2$$

$$= 124k + 440k^2 - (3.66)^2$$

$$12.4 + 4.4 - 13.3956$$

$$= 16.8 - 13.3956$$

$$= 3.4044$$

a) A random variable X has the following probability function

x	-3	-2	-1	0	1	2	3
$P(X=x)$	k	0.1	k	0.2	$2k$	0.4	$2k$

Find k , mean & variance

w.k.t $f(x) = P(X=x)$ satisfies

$$(i) f(x) \geq 0$$

$$(ii) \sum f(x) = 1$$

$$\Rightarrow k > 0$$

$$\sum f(x) = k + 0.1 + k + 0.2 + 2k + 0.4 + 2k$$

$$= 6k + 0.7 = 1$$

$$k = \frac{0.3}{6} = \frac{0.1}{2} = \frac{1}{20} = \underline{\underline{0.05}}$$

$$\text{Mean} = \sum x f(x)$$

$$= (-3)k - 2(0.1) - k + 0 + 2k + 0.4 \times 2 + 6k$$

$$4(0.05) + 0.6$$

$$= 0.2 + 0.6$$

$$= \underline{\underline{0.8}}$$

$$\text{Variance} = \sum x^2 f(x) - \mu^2$$

$$= 9K + 4(0.1) + K + 0 + 2K + 4(0.4) + 9(2K)$$

$$= 30K + 2 - (0.8)^2$$

$$= 31.5 + 2 - (0.8)^2$$

$$= 3.5 - 0.64$$

$$= \underline{\underline{2.86}}$$

10-09-25

Theorem 1: If X is a random variable & a and b are constants, then, (i) $E(ax+b) = aE(x) + b$

$$(ii) \text{Var}(ax+b) = a^2 \text{Var}(x)$$

[If x is discrete, use summations]

[If x is continuous, use integrations]

Proof:

(i) We know that X is a random variable, a & b are constants, then $ax+b$ is also a random variable.

$$\text{w.k.t. } E(x) = \sum_x x f(x)$$

$$\text{Now } E(ax+b) = \sum_{ax+b} (ax+b) f(ax+b)$$

$$= \sum_x (ax+b) f(x)$$

$$= a \sum x f(x) + b \sum f(x)$$

$$= a \sum x f(x) + b(1) \because \sum f(x) = 1$$

$$= a E(x) + b \because \sum x f(x) = E(x)$$

$$\therefore \boxed{E(ax+b) = aE(x) + b} \quad \text{--- (1)}$$

NOTE: For Continuous R.V replace summation with

Integration, $\left[\int_{-\infty}^{\infty} x f(x) \right]$

(ii) we know that

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\begin{aligned}\Rightarrow \text{Var}(aX+b) &= E((aX+b)^2) - (E(aX+b))^2 \\ &= E(a^2X^2 + b^2 + 2abX) - (aE(X) + b)^2 \quad [\text{from (i)}]\end{aligned}$$

$$= a^2 E(X^2) + b^2 + 2ab E(X)$$

$$- (a^2 (E(X))^2 + b^2 + 2ab E(X))$$

$$= a^2 E(X^2) + \cancel{b^2} + 2ab \cancel{E(X)}$$

$$- a^2 (E(X))^2 - \cancel{b^2} - 2ab \cancel{E(X)}$$

$$= a^2 [E(X^2) - (E(X))^2]$$

$$\boxed{\text{Var}(aX+b) = a^2 \text{Var}(X)}$$

Q1 A sample of 4 items is selected at random from a box containing 12 items of which 3 are defective.

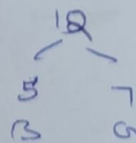
Find the expected number E of defective items.

Let X represents defectives.

Among the 4 out of 12.

X can take $\{0, 1, 2, 3, 4\}$
the values

$X=0 \Rightarrow$ No defectives.



$$P(X=0) = \frac{{}^7C_4}{{}^{12}C_4} = \frac{7 \times 6 \times 5 \times 4}{\cancel{6} \times \cancel{12} \times 11 \times 10 \times 9} = \frac{7}{99}$$

$$X=1 \Rightarrow P(X=1) \Rightarrow \frac{{}^5C_1 \cdot {}^7C_3}{{}^{12}C_4} = \frac{5 \times 35}{495} = \frac{35}{99}$$

$$X=2 \Rightarrow P(X=2) \Rightarrow \frac{{}^5C_2 \cdot {}^7C_2}{{}^{12}C_4} = \frac{10 \times 7 \times 6}{2 \times 495} = \frac{42}{99}$$

$$X=3 \Rightarrow P(X=3) \Rightarrow \frac{{}^5C_3 \cdot {}^7C_1}{{}^{12}C_4} = \frac{10 \times 7}{495} = \frac{14}{99}$$

$$X=4 \Rightarrow P(X=4) \Rightarrow \frac{{}^5C_4 \cdot {}^7C_0}{{}^{12}C_4} = \frac{5}{495} = \frac{1}{99}$$

$$\text{sum} \rightarrow \frac{7+35+42+14+1}{99} = \frac{99}{99} = 1 \quad \{\text{verified}\}$$

$X=x$	0	1	2	3	4
$P(X=x)$	$\frac{7}{99}$	$\frac{35}{99}$	$\frac{42}{99}$	$\frac{14}{99}$	$\frac{1}{99}$

$$\text{Expected no. of defects} = E(X) = \sum x f(x)$$

$$= \sum_{x=0}^4 x P(X=x)$$

$$= 0 + \frac{35}{99} + \frac{84}{99} + \frac{42}{99} + \frac{4}{99}$$

$$= \frac{166}{99} = 1.667$$

Q) Find the mean and variance of the uniform probability distribution given by $f(x) = \frac{1}{N}$, for $x = 1, 2, 3, \dots, N$.

sol Probability distribution is given by

$X=x$	1	2	3	4	...	N
$P(X=x)$ $f(x)$	$\frac{1}{N}$	$\frac{1}{N}$	$\frac{1}{N}$	$\frac{1}{N}$	...	$\frac{1}{N}$

$$\text{Mean} = E(X) = \sum_{x=1}^N x f(x)$$

$$= \frac{1}{N} + \frac{2}{N} + \dots + \frac{N}{N} = \frac{N(N+1)}{2 \cdot N} = \frac{N+1}{2} //$$

$$\text{Variance} = \sum x^2 f(x) - \mu^2$$

$$= 1^2\left(\frac{1}{N}\right) + 2^2\left(\frac{1}{N}\right) + \dots + N^2\left(\frac{1}{N}\right) - \left(\frac{N+1}{2}\right)^2$$

$$= \frac{N(N+1)(2N+1)}{6} \cdot \frac{1}{N} - \frac{(N+1)^2}{4}$$

$$= \left(\frac{N+1}{2}\right) \left[\frac{2N+1}{3} - \frac{N+1}{2} \right]$$

$$= \left(\frac{N+1}{2}\right) \left[\frac{4N+2-3N-3}{6} \right]$$

$$= \frac{N^2-1}{12} //$$

Q) A player tosses 2 fair coins. He wins ₹100 if head appears, ₹200 if two heads appear. On the other hand, he loses ₹500 if no head appears. Determine the expected value E of the game, is the game favorable to the player?

Let x represents no. of heads.

x can take the values 0, 1, 2

$\left\{ \begin{array}{l} H H \\ H T \\ T H \\ T T \end{array} \right\}$

$x=x$	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$E(x) = ₹ \frac{1}{4} \times (-500) + \frac{2}{4} \times (100) + \frac{1}{4} \times (200)$$

$$= \frac{-500 + 200 + 200}{4} = \frac{-100}{4}$$

$$= -25$$

$E < 0$, it is not favorable

Q1) Two dice are thrown. Let X assign to each point (a, b) in S , the maximum of its numbers, i.e., $X(a, b) = \max(a, b)$. Find the probability distribution if X is a random variable with $X(S) = \{1, 2, 3, 4, 5, 6\}$. Also find mean & variance of the distribution.

Total no. of cases when two dice are thrown

$$S = 6 \times 6 = 36 = \begin{pmatrix} (1,1), (1,2), \dots, (1,6) \\ \vdots \\ (6,1), (6,2), \dots, (6,6) \end{pmatrix}$$

Given $X(a, b) = \max(a, b)$.

Then $(X=1) \Rightarrow$ max of two no's is 1.

$$P(X=1) = \frac{1}{36} \{(1,1)\}$$

$$P(X=2) = \frac{3}{36} \{(1,2), (2,1), (2,2)\}$$

$$P(X=3) = \frac{5}{36} \{(1,3), (3,1), (3,2), (2,3), (3,3)\}$$

$$P(X=4) = \frac{7}{36} \{(1,4), (4,1), (2,4), (4,2), (3,4), (4,3), (4,4)\}$$

$$P(X=5) = \frac{9}{36} \{(1,5), (5,1), (2,5), (5,2), (3,5), (5,3), (4,5), (5,4), (5,5)\}$$

$$P(X=6) = \frac{11}{36}$$

$X=x$	1	2	3	4	5	6
$P(X=x)$	$1/36$	$3/36$	$5/36$	$7/36$	$9/36$	$11/36$

$$\mu = E(X) = \sum_{x=1}^6 x \cdot P(X=x) = \frac{1}{36} + \frac{6}{36} + \frac{15}{36} + \frac{28}{36} + \frac{45}{36} + \frac{66}{36} = 4.472$$

$$\Rightarrow \text{variance} = \sum x^2 f(x) - \mu^2$$

$$= \frac{1}{36} + \frac{4 \cdot 3}{36} + \frac{9 \cdot 5}{36} + \frac{16 \cdot 7}{36} + \frac{25 \cdot 9}{36} + \frac{36 \cdot 11}{36}$$

$$= \frac{1 + 12 + 45 + 112 + 225 + 396}{36} - (4.72)^2$$

$$= 21.972 - 19.998$$

$$= \underline{\underline{1.973}}$$

Q1 Let X denotes no. of heads in a single toss of 4 fair coins

(i) Determine $P(X < 2)$.

(ii) $P(1 < X \leq 3)$

X represents no. of heads

X can take the values 0, 1, 2, 3, 4.

Sample space $S = 16 = \left\{ \begin{array}{l} (T, T, T, T) \\ (T, H, T, T) \\ (H, T, T, T) \\ (H, H, T, T) \end{array} \right\} = (2^4)$

Total outcomes $= 2^4 = 16$

$X = n$	0	1	2	3	4
$P(X = n)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$$\begin{aligned} \text{i) } P(X < 2) &= P(X=0) + P(X=1) \\ &= \frac{1}{16} + \frac{4}{16} = \frac{5}{16} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(1 < X \leq 3) &= P(X=2) + P(X=3) \\ &= \frac{6}{16} + \frac{4}{16} = \frac{10}{16} = \frac{5}{8} \end{aligned}$$

Q1 A fair coin is tossed until a head or tails occurs. Find the expected number E of tosses of the coin.

Let X represents number of tosses of the coin.

$$P(1) = P(H) = \frac{1}{2}$$

$$P(2) = P(TH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(3) = P(TTH) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$P(4) = P(TTTH) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$$

$$P(5) = P(TTTTH) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{32}$$

$$P(6) = P(TTTTT) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{32} \quad \left. \begin{array}{l} P(5) \\ P(6) \end{array} \right\} + = P(5)$$

$\therefore X$ takes the values 1, 2, 3, 4, 5, 6

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{32}$

$$E(X) = \sum_{x=1}^6 x \cdot f(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{16} + \frac{6}{32}$$

$$= \frac{16 + 16 + 12 + 8 + 11}{32}$$

$$= \frac{63}{32} = 1.968$$

$$= 1 + \frac{15}{16} = \frac{31}{16} = 1.9375$$

Q1 A random variable X is defined as sum of the no. of faces when two dice are thrown. Find $\text{mean}(X)$.

X represents sum of faces.

X takes the values 2, 3, ..., 12

total cases = 36

$X = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$

$$P(X=2) = \frac{1}{36}$$

$$P(X=8) = \frac{5}{36}$$

$$P(X=3) = \frac{2}{36}$$

$$P(X=9) = \frac{4}{36}$$

$$P(X=4) = \frac{3}{36}$$

$$P(X=10) = \frac{3}{36}$$

$$P(X=5) = \frac{4}{36}$$

$$P(X=11) = \frac{2}{36}$$

$$P(X=6) = \frac{5}{36}$$

$$P(X=12) = \frac{1}{36}$$

$$P(X=7) = \frac{6}{36}$$

$X = x$	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$\begin{aligned} \mu = E(X) &= \frac{2}{36} + \frac{6}{36} + \frac{12}{36} + \frac{20}{36} + \frac{30}{36} + \frac{42}{36} \\ &\quad + \frac{40}{36} + \frac{36}{36} + \frac{30}{36} + \frac{22}{36} + \frac{12}{36} \\ &= \frac{252}{36} = 7 \end{aligned}$$

2) A continuous random variable has the probability density $f(x)$ $f(x) = \begin{cases} kxe^{-\lambda x}, & \text{for } x \geq 0, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$

Determine E, mean & variance

We know that $f(x)$ [Prob. density $f(x)$] satisfies the following properties.

$$(i) f(x) \geq 0 \Rightarrow kxe^{-\lambda x} \geq 0 \Rightarrow k > 0$$

$$(ii) \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} kxe^{-\lambda x} dx = 1$$

$$k \int_{-\infty}^{\infty} xe^{-\lambda x} dx \Rightarrow k \left[\frac{xe^{-\lambda x}}{-\lambda} - \frac{e^{-\lambda x}}{\lambda^2} \right]_{-\infty}^{\infty} = 1$$

$$\Rightarrow k \left[\frac{-e^{-\lambda x}}{\lambda} \left[x + \frac{1}{\lambda} \right] \right]_{-\infty}^{\infty} = 1$$

$$\Rightarrow -k \left[\frac{x}{\lambda e^{\lambda x}} + \frac{e^{-\lambda x}}{\lambda^2} \right]_{-\infty}^{\infty} = 1$$

$$\Rightarrow -k \left[\left[0 + 0 \right] - \left[0 + \frac{1}{\lambda^2} \right] \right] = 1$$

$$\Rightarrow \frac{k}{\lambda^2} = 1$$

$$k = \lambda^2$$

$$E(x) = \mu = \int_0^{\infty} xf(x) dx = \int_0^{\infty} x \lambda^2 \cdot xe^{-\lambda x} dx = \int_0^{\infty} \lambda^2 x^2 e^{-\lambda x} dx$$

$$= \lambda^2 \left[\frac{x^2 e^{-\lambda x}}{-\lambda} - \frac{2xe^{-\lambda x}}{\lambda^2} + \frac{2e^{-\lambda x}}{-\lambda^3} \right]_0^{\infty}$$

$$= \lambda^2 \left[(0) + \left(\frac{2}{+\lambda^3} \right) \right]$$

$$= \frac{2}{\lambda} //$$

$$\text{Variance} \Rightarrow \sigma^2 = \int_0^{\infty} x^2 f(x) dx - \mu^2$$

$$= \int_0^{\infty} x^2 \cdot \lambda^2 e^{-\lambda x} dx - \frac{4}{\lambda^2}$$

$$= \lambda^2 \int_0^{\infty} x^3 e^{-\lambda x} dx$$

$$= \lambda^2 \left[\frac{x^3 e^{-\lambda x}}{-\lambda} - \frac{3x^2 e^{-\lambda x}}{\lambda^2} + \frac{6x e^{-\lambda x}}{-\lambda^3} - \frac{6e^{-\lambda x}}{\lambda^4} \right]_0^{\infty}$$

$$= \lambda^2 \left[(0) - \left(\frac{-6e^{-\lambda x}}{\lambda^4} \right) \right] = \frac{4}{\lambda^2}$$

$$= \frac{+6}{\lambda^2} - \frac{4}{\lambda^2} = \frac{2}{\lambda^2} = \sigma^2$$

$$\text{Standard deviation} \Rightarrow \sqrt{\frac{2}{\lambda^2}} = \frac{\sqrt{2}}{\lambda}$$

Q) The probability density function $f(x)$ is given by

$$f(x) = c e^{-|x|}, \quad -\infty < x < \infty. \text{ Show that } c = 1/2,$$

2) Find mean and variance of that distribution.

Also find the probability that the variate lies b/w 0 and 4. $\{P(0 < x < 4)\}$

We know that (continuous variable) $f(x)$ satisfies the following;

$$i) f(x) \geq 0 \Rightarrow c > 0 \quad \because e^{-|x|} > 0$$

$$ii) \int_{-\infty}^{\infty} f(x) dx = 1$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\Rightarrow \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^0 ce^x dx + \int_0^{\infty} ce^{-x} dx = 1$$

$$\Rightarrow c[e^x]_{-\infty}^0 + c[-e^{-x}]_0^{\infty} = 1$$

$$\Rightarrow c[1-0] + c[0-(-1)] = 1$$

$$\Rightarrow c+c=1$$

$$\Rightarrow 2c=1$$

$$\Rightarrow \boxed{c=1/2} \text{ Hence proved}$$

$$\mu = \text{Mean} = \int_{-\infty}^{\infty} xf(x) dx = \int_{-\infty}^0 xf(x) dx + \int_0^{\infty} xf(x) dx$$

$$= \int_{-\infty}^0 x \cdot \frac{1}{2} \cdot e^x dx + \int_0^{\infty} x \cdot \frac{1}{2} \cdot e^{-x} dx$$

$$= \left[\frac{xe^x - e^x}{2} \right]_{-\infty}^0 + \left[\frac{-xe^{-x} - e^{-x}}{2} \right]_0^{\infty}$$

$$= \left[\frac{0-1}{2} \right] - [0] + \left[\frac{0-0}{2} \right] - \left[0 - \frac{1}{2} \right]$$

$$= \frac{-1}{2} + \frac{1}{2} = \underline{\underline{0}}$$

$$\therefore \mu = 0$$

$$\text{Variance} = \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx + \int_0^{\infty} x^2 f(x) dx - \mu^2$$

$$= \frac{1}{2} \int_{-\infty}^0 x^2 e^x dx + \frac{1}{2} \int_0^{\infty} x^2 e^{-x} dx$$

$$= \left[\frac{x^2 e^x - 2x e^x + 2e^x}{2} \right]_0^0 + \left[\frac{x^2 e^{-x} - 2x e^{-x} - 2e^{-x}}{2} \right]_0^0$$

$$\left[\left[\frac{0-0+2}{2} \right] - \left[\frac{0-0-0}{2} \right] \right] + \left[\left[\frac{0-0-0}{2} \right] - \left[\frac{0-0-2}{2} \right] \right]$$

$$= 1 + 1$$

$$\sigma^2 = 2$$

$$\therefore \text{S.D} = \sigma = \sqrt{2}$$

$$P(0 < X < 4) = \int_0^4 f(x) dx = \int_0^4 \frac{e^x}{2} dx = \left[\frac{e^x}{2} \right]_0^4 = \left[\frac{e^4}{2} - \frac{1}{2} \right]$$

$$\therefore P(0 < X < 4) = \frac{e^4 - 1}{2}$$

Q] A continuous random variable 'X' has the distribution

$$f(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ k(x-1)^4 & \text{if } 1 \leq x \leq 3 \\ 0 & \text{if } x > 3 \end{cases}$$

find i) $f(x)$ ii) k iii) Mean

We know that $\frac{d}{dx} F(x) = f(x)$

$$f(x) = \frac{d}{dx} F(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ 4k(x-1)^3 & \text{if } 1 \leq x \leq 3 \\ 0 & \text{if } x > 3 \end{cases}$$

(ii) $\therefore f(x)$ is PDF, it satisfies

$$1. f(x) \geq 0 \Rightarrow 4k(x-1)^3 \geq 0 \quad [\because (x-1)^3 > 0 \text{ in this interval}], k > 0.$$

$$2. \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_{-\infty}^1 \underbrace{f(x)}_0 dx + \int_1^3 4k(x-1)^3 dx + \int_3^{\infty} \underbrace{f(x)}_0 dx = 1$$

$$\Rightarrow \int_1^3 4k(x-1)^3 dx = 1 \Rightarrow 4k \left[\frac{(x-1)^4}{4} \right]_1^3 = 1$$

$$\rightarrow 4k \left[\frac{16}{4} - 0 \right] = 1$$

$$16k = 1 \Rightarrow \boxed{k = 1/16}$$

$$(iii) \text{ Mean} = \mu = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^1 x \cdot 0 dx + \frac{1}{16} \int_1^3 x(x-1)^3 dx + \int_3^{\infty} x \cdot 0 dx$$

$$\Rightarrow \frac{1}{16} \int_1^3 x [x^3 - 1 - 3x^2 + 3x] dx$$

$$\Rightarrow \frac{1}{16} \int_1^3 (x^4 - 3x^3 + 3x^2 - x) dx$$

$$\Rightarrow \frac{1}{16} \left[\frac{x^5}{5} - \frac{3x^4}{4} + x^3 - \frac{x^2}{2} \right]_1^3$$

$$\Rightarrow \frac{1}{16} \left[\left(\frac{243}{5} - \frac{243}{4} + 27 - \frac{9}{2} \right) - \left(\frac{1}{5} - \frac{3}{4} + 1 - \frac{1}{2} \right) \right]$$

$$\Rightarrow \frac{1}{16} \left[\left(\frac{-243}{20} + \frac{45}{2} \right) - \left(\frac{7}{10} - \frac{3}{4} \right) \right]$$

$$\Rightarrow \frac{1}{16} \left[\frac{-243 + 450}{20} + \frac{2}{20} \right]$$

$$\Rightarrow \frac{1}{16} \left[\frac{209}{20} \right]$$

12/09/25

2) The trouble shooting capability of an IC (circuit) chip in a circuit is 'x' whose distribution is

$$\text{given by } f(x) = \begin{cases} 0 & \text{for } x \leq 3 \\ 1 - \frac{9}{x^2} & \text{for } x > 3. \end{cases} \text{ where } x$$

denotes no. of years. Find the probability that the IC chip will work properly:

- (i) Less than 8 years
- (ii) Beyond 8 years
- (iii) Between 5 to 7 years
- (iv) Anywhere from 2 to 5 years.

We know that $f(x) = \frac{d}{dx} (F(x))$

$$\rightarrow f(x) = \begin{cases} 0, & x \leq 3 \\ \frac{+18}{x^3}, & x > 3 \end{cases}$$

$$\begin{aligned} \text{(i) } P(X < 8) &= \int_{-\infty}^8 f(x) dx = \int_{-\infty}^3 f(x) dx + \int_3^8 f(x) dx \\ &= \int_{-\infty}^3 0 dx + \int_3^8 \frac{-18}{x^3} dx \\ &= - \left[\frac{+18}{+2 \cdot x^2} \right]_3^8 \\ &= - \left(\frac{9}{64} - \frac{9}{9} \right) = \frac{+55}{64} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(X < 8) &= P(X \leq 8) = \underbrace{0 + 0 + 0 + P(X=3) + P(X=4) + P(X=5)}_{+ P(X=6) + P(X=7) + P(X=8)} \\ &= 1 - \frac{9}{16} + 1 - \frac{9}{25} + 1 - \frac{9}{36} + 1 - \frac{9}{49} + 1 - \frac{9}{64} \end{aligned}$$

$$\begin{aligned} P(X \leq 8) &= F(8) \\ &= 1 - \frac{9}{64} = \frac{55}{64} \end{aligned}$$

$$\text{(i) } P(X > 8) = 1 - P(X \leq 8) = 1 - \frac{55}{64} = \frac{9}{64}$$

$$\begin{aligned} \text{(ii) } P(5 < X < 7) &= F(7) - F(5) = 1 - \frac{9}{49} + 1 - \frac{9}{25} \\ &= 2 - 9 \left[\frac{1}{49} + \frac{1}{25} \right] = \end{aligned}$$

$$(iv) P(2 < X < 5) = P(5) - P(2)$$

$$= 1 - \frac{9}{25} - 0 = \frac{16}{25} //$$

Q1 Find the density funcⁿ of random variable X is

$$f(x) = \begin{cases} \frac{1}{2} \sin x & 0 \leq x \leq \pi \\ 0 & \text{otherwise} \end{cases} \quad \text{Find Mean, median,}$$

mode of the distribution also find the probability
 $0 \leq \pi/2 \in P(0 < x < \pi/2)$

Mode

w.k.T.

mean is given by $\mu = \int_{-\infty}^{\infty} x f(x) \cdot dx$

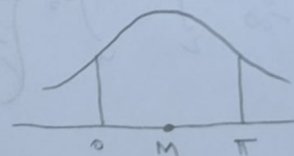
$$\Rightarrow \mu = \int_{-\infty}^0 x f(x) dx + \int_0^{\pi} x f(x) dx + \int_{\pi}^{\infty} x f(x) dx$$

$$= \int_{-\infty}^0 0 dx + \int_0^{\pi} \frac{x}{2} \sin x dx + \int_{\pi}^{\infty} 0 dx$$

$$= 0 + \left[-\frac{x}{2} \cos x \right]_0^{\pi} + 0 + \left[+ \frac{\sin x}{2} \right]_0^{\pi}$$

$$\Rightarrow \frac{-\pi}{2} \left[\frac{\pi - 0}{2} \right] + [0 - 0]$$

$$\Rightarrow \frac{\pi}{2} = \frac{\pi}{2}$$



Median

Let 'M' be the median. Then, according to the definition

of median, we get $\int_0^M f(x) dx = \int_M^{\pi} f(x) dx = \frac{1}{2}$

$$\text{But } \int_0^M f(x) dx + \int_M^{\pi} f(x) dx = 1 \Rightarrow \text{each is } 1/2.$$

Consider any one of the two.

$$\int_0^M f(x) dx = \frac{1}{2} \Rightarrow \int_0^M \frac{1}{2} \sin x dx = \frac{1}{2}$$

$$(-\cos x)_0^M = 1 \Rightarrow x - \cos M = x$$

$$\Rightarrow \cos M = 0$$

$$\Rightarrow M = \pi/2$$

Mode: It is the value of x for which $f(x)$ is maximum.

Given, $f(x) = \frac{1}{2} \sin x$, $0 \leq x \leq \pi$

$$f'(x) = \frac{1}{2} \cos x$$

$$f'(x) = 0 \rightarrow \cos x = 0 \rightarrow x = \pi/2$$

$$f''(x) = -\frac{1}{2} \sin x$$

max value $f''(\pi/2) = -\frac{1}{2} \sin \pi/2 = -\frac{1}{2} < 0$

$f(x)$ is max at $x = \pi/2$

$\therefore \text{Mode} = \pi/2$

Q) Let the function defined by $f(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{18}(2x+3), & 2 \leq x \leq 4 \\ 0, & x > 4 \end{cases}$ a PDF?

Find the prob. that a variate having $f(x)$ is a PDF if $\Rightarrow$ PDF will fall in interval $P(2 \leq x \leq 3)$

(i) $f(x) \geq 0 \rightarrow$ Yes. In $2 \leq x \leq 4$, $\frac{1}{18}(2x+3) > 0$ ✓

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$ Clearly, $f(x) \geq 0 \forall x$

$$= \int_{-\infty}^2 0 + \int_2^4 \frac{1}{18}(2x+3) + \int_4^{\infty} 0$$

$$= \frac{1}{18} \left[\frac{(2x+3)^2}{2} \right]_2^4 = \frac{1}{36} [121 - 49]$$

$$= \frac{1}{18} \left[x^2 + 3x \right]_2^4 = \frac{1}{18} [16 + 12 - 4 - 6] = \frac{18}{18} = 1$$

$\therefore f(x)$ is a PDF

$P(2 \leq x \leq 3) = \frac{1}{18} \int_2^3 (2x+3) = \frac{1}{18} \left(\frac{x^2}{2} + 3x \right)_2^3$

$$= \frac{1}{18} [9 + 9 - 4 - 6] = \frac{8}{18} = \frac{4}{9}$$

Theorem: (Joint probability)

If X & Y are independent random variables, then

$$E(XY) = E(X) \cdot E(Y)$$

Proof: W.K.T if X & Y are random variables, then XY is also a random variable.

$$E(X) = \sum x f(x), \quad E(Y) = \sum y f(y)$$

$$\text{Then } E(XY) = \sum_x \sum_y xy f(x, y) \quad P(X \cap Y) = P(X) \cdot P(Y)$$

$$= \sum_x \sum_y xy f(x) f(y) \quad \text{if } X \text{ \& } Y \text{ are independent}$$

$$= \sum_x x f(x) \cdot \sum_y y f(y)$$

$$= E(X) \cdot E(Y)$$

$$\therefore E(XY) = E(X) \cdot E(Y)$$

(Marginal probabilities)

Theorem: If X and Y are the random variables, then

$$E(X+Y) = E(X) + E(Y)$$

Proof: W.K.T if X and Y are random variables, then $X+Y$ is also a random variable.

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx, \quad E(Y) = \int_{-\infty}^{\infty} y f(y) dy$$

$$\Rightarrow E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f(x,y) dy \right] dx + \int_{-\infty}^{\infty} y \left[\int_{-\infty}^{\infty} f(x,y) dx \right] dy$$

$$= \int_{-\infty}^{\infty} x f(x) dx + \int_{-\infty}^{\infty} y f(y) dy$$

$$= E(X) + E(Y) \quad \text{Hence proved}$$

If X is a C.R.V with PDF given by

$$Q) f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2-x & \text{if } 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E(25x^2 + 30x - 5)$

$$\Rightarrow \text{W.K.T. } E(25x^2 + 30x - 5) =$$

$$= E(25x^2 + 25x - 5x)$$

$$= 25E(x^2) + 30E(x) - 5$$

$$E(x^2) = [E(x)]^2$$

2. given $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 2-x & \text{if } 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$

$$\Rightarrow E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^0 0 + \int_0^1 x dx + \int_1^2 (2-x) dx + \int_2^{\infty} 0 dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^2$$

$$= \frac{1}{2} + (4 - 2) - (2 - \frac{1}{2})$$

$$= \frac{1}{2} + 2 - 2 + \frac{1}{2} = 1$$

$$E(x) = 1$$

$$\begin{aligned}
 E(x^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^0 0 dx + \int_0^1 x^2 \cdot x dx + \int_1^2 x^2 (2-x) dx + \int_2^{\infty} 0 dx \\
 &= 0 + \left[\frac{x^4}{4} \right]_0^1 + \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_1^2 + 0 \\
 &= \frac{1}{4} + \frac{16}{3} - 4 - \frac{2}{3} + \frac{1}{4} \\
 &= \frac{1}{2} + \frac{14}{3} - 4 \\
 &= \frac{3+28-24}{6} = \frac{7}{6} //
 \end{aligned}$$

$$\therefore E(25x^2 + 30x - 5) = 25 \left(\frac{7}{6} \right) + 30(1) - 5$$

$$= \frac{175}{6} + 25$$

$$= \frac{175 + 150}{6}$$

$$= \frac{325}{6} //$$

COVARIANCE:

Covariance is a statistical measure that tells us how both the variables X and Y change together. If both increase or decrease, then the covariance is positive.

- If one increases and other decreases, then the covariance is negative.
- If X and Y are independent, then the covariance tends towards zero (0).

Let $(\overset{x}{x}, \overset{y}{y})$ be a two dimensional or a bivariate random variable. Then, the covariance of random variables X and Y is denoted by $\text{cov}(X, Y)$ and is defined as

$$\text{COV}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E[XY - XE(Y) - YE(X) + E(X) \cdot E(Y)]$$

$$= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y)$$

$$\text{COV}(X, Y) = E(XY) - E(X) \cdot E(Y) \quad \text{--- (1)}$$

NOTE: We know that

If X and Y are independent, $E(XY) = E(X)E(Y)$

From (1), $\text{COV}(X, Y) = E(X) \cdot E(Y) - E(X)E(Y)$

$$\therefore \boxed{\text{COV}(X, Y) = 0}$$

But converse need not be true.

NOTE 1: $\text{COV}(aX, bY) = ab \text{COV}(X, Y)$

$$\text{COV}(a+X, b+Y) = \text{COV}(X, Y)$$

Q For the following joint probability distribution, find $\text{COV}(X, Y)$

$X/Y \rightarrow$	1	2	3	P_i
0	$2/12$	$1/12$	$1/12$	$4/12$
1	$1/12$	$1/12$	$2/12$	$4/12$
2	$3/12$	$1/12$	0	$4/12$
P_j	$6/12$	$3/12$	$3/12$	1

w.k.t. $E(X) = \sum x f(x) = \sum x P_i = 0 + \frac{4}{12} + \frac{8}{12} = 1$

$$E(Y) = \sum y f(y) = \sum y P_j = \frac{6}{12} + \frac{6}{12} + \frac{9}{12} = \frac{21}{12} = \frac{7}{4}$$

$$E(XY) = \sum \sum xy f(x, y) = \sum \sum xy P_{ij}$$

$$= 0 \cdot 1 \cdot \frac{2}{12} + 0 \cdot 2 \cdot \frac{1}{12} + 0 \cdot 3 \cdot \frac{1}{12} + 1 \cdot 1 \cdot \frac{1}{12} + 1 \cdot 2 \cdot \frac{1}{12} + 1 \cdot 3 \cdot \frac{2}{12}$$

$$+ 2 \cdot 1 \cdot \frac{3}{12} + 2 \cdot 2 \cdot \frac{1}{12} + 2 \cdot 3 \cdot 0$$

$$= 0 + \frac{1}{12} + \frac{2}{12} + \frac{6}{12} + \frac{6}{12} + \frac{4}{12}$$

$$= 1 + \frac{9}{12} = \frac{19}{12} = \left\{ \frac{7}{4} \right\}$$

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= \frac{19}{12} - 1 \cdot \frac{7}{4}$$

$$= \frac{19}{12} - \frac{21}{12} = -\frac{2}{12} = -\frac{1}{6} //$$

18-09-2025

$$\begin{aligned} * \text{cov}(x, y) &= E((X - \bar{X})(Y - \bar{Y})) \\ &= \frac{\sum (x - \bar{x})(y - \bar{y})}{n} \end{aligned}$$

$$* \text{corr } r = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y}$$

Q Find covariance b/w heights of fathers & heights of sons.
 { Draw a table with 9 columns and 2 rows }

Ht of fathers	65	66	67	67	68	69	71	73
Ht of sons	67	68	64	68	72	70	69	70

small data
 $\Rightarrow (n-1)$

Let x, y be heights of father & son resp.

$$\bar{x} = \frac{\sum x}{n} = \frac{65 + 66 + 67 + \dots + 73}{8} = 68.25$$

$$\bar{y} = \frac{\sum y}{n} = \frac{67 + 68 + 64 + 68 + 72 + 70 + 69 + 70}{8} = 68.5$$

$$\Rightarrow \bar{x} = 68.25, \quad \bar{y} = 68.5$$

x	y	$x = x - \bar{x}$	$y = y - \bar{y}$	xy	$x^2(x - \bar{x})^2$	$y^2(y - \bar{y})^2$
65	67	-3.25	-1.5	4.875	10.5625	2.25
66	68	-2.25	-0.5	1.125	5.0625	0.25
67	64	-1.25	-4.5	5.625	1.5625	20.25
67	68	-1.25	-0.5	0.625	1.5625	0.25
68	72	-0.25	+3.5	-0.875	0.0625	12.25
69	70	0.75	1.5	1.125	0.5625	2.25
71	69	2.75	0.5	1.375	7.5625	0.25
73	70	4.75	1.5	7.125	22.5625	2.25

$$\sum(x - \bar{x})(y - \bar{y})$$

$$= 21$$

$$\sum(x - \bar{x})^2 = 49.5$$

$$\sum(y - \bar{y})^2 = 40$$

$$\text{cov}(x, y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n} = \frac{21}{8} = 2.625$$

$$\sum(x - \bar{x})^2 = 49.5, \quad \sum(y - \bar{y})^2 = 40$$

$$\text{corr } r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y} \quad \sigma_x = \sqrt{\frac{49.5}{8}}, \quad \sigma_y = \sqrt{\frac{40}{8}}$$

$$= 2.487$$

$$= 2.236$$

$$= \frac{2.625}{(2.487)(2.236)} = 0.472$$

NOTE: If $-1 \leq r \leq 1$, r is closer to -1 or closer to 1, then the strength of relation is said to be strong. Covariance doesn't lie b/w $(-1, 1)$, so we use corr to get in that range.

Chebyshev's inequality

Let X be a random variable with mean μ and variance σ^2 , then Chebyshev's inequality is given by

$$P(|X - \mu| > k) \leq \frac{\sigma^2}{k^2}$$

$$\text{or } P(|X - \mu| < k) \geq 1 - \frac{\sigma^2}{k^2} \Rightarrow P(\mu - k < X < \mu + k) \geq 1 - \frac{\sigma^2}{k^2}$$

$$\text{or } P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

8] If the Chebyshev's inequality for the random variable X is $P(-2 < X < 8) \geq \frac{21}{25}$, Find $E(X)$ & $\text{Var}(X)$

From the Chebyshev's inequality, w.k.t

$$P(\mu - k < X < \mu + k) \geq 1 - \frac{\sigma^2}{k^2}$$

$$\Rightarrow \mu - k = -2$$

$$\mu + k = 8$$

$$2\mu = 6 \Rightarrow \mu = 3$$

$$k = 5$$

$$\text{And } 1 - \frac{\sigma^2}{k^2} = \frac{21}{25} \Rightarrow 1 - \frac{\sigma^2}{25} = \frac{21}{25}$$

$$\frac{4}{25} = \frac{\sigma^2}{25}$$

$$\Rightarrow \sigma^2 = 4$$

$$\therefore \text{Mean} = \mu = 3$$

$$\text{Variance} = \sigma^2 = 4$$

Q) The number of components manufactured in a factory during a one month period is a random variable with $\mu = 600$, $\sigma^2 = 100$. What is the probability that the production will be b/w 500, 700 over a month.

To find, $P(500 < X < 700)$.

From Chebyshev's inequality, w.k.t

$$P(\mu - k < X < \mu + k) \geq 1 - \frac{\sigma^2}{k^2}$$

Comparing

$$\mu - k = 500 \Rightarrow \mu = 600;$$

$$600 - k = 500 \Rightarrow k = 100$$

$$\therefore P(500 < X < 700) \geq 1 - \frac{100}{(100)^2}$$

$$\geq \frac{99}{100}$$

$$P(500 < X < 700) \geq 0.99$$

Binomial Distribution:

Consider a random experiment. If this experiment is repeated n times with only 2 outcomes (mutually disjoint outcomes), i.e., success and failure, then the trials of this experiment are called Bernoulli's trials (or ~~experiment~~) where occurrence is called success, non occurrence is called failure.

Probability of Success is denoted with p ,

Probability of Failure is denoted with q ,

$$\text{Then } p + q = 1.$$

The probability of getting x successes in some specified order is given by $\rightarrow$ ①

w.k.t getting x successes out of n trials can be done in nCx ways. ① $\Rightarrow P(X=x) = nCx p^x q^{n-x}$ $x \geq 0, x \leq n$
 $\rightarrow x = 0, 1, \dots, n$

→ This is called Binomial Distribution as $(x+1)^n$ term of $(q+p)^n$ matches with $b(x, n, p) = P(X=x) = {}^n C_x p^x q^{n-x}$

→ Binomial distribution is a discrete distribution, and the variable which follows Binomial distribution is called binomial variate.

$$B(x, n, p) = \begin{cases} {}^n C_x p^x q^{n-x}, & x \geq 0, x \leq n \\ 0 & \text{otherwise} \end{cases}$$

$x = 0, 1, \dots, n$

Here, n and p are called parameters of Binomial distribution.

→ The following are the conditions to be satisfied for applying Binomial distribution.

- 1) No. of trials should be finite.
- 2) Trials should be Independent of each other.
- 3) Each trial should result in only 2 mutually disjoint outcomes, i.e.; success and failure.
- 4) Probability of success should be same in all the trials.

Mean of the binomial distribution

∵ Binomial is a discrete distribution, mean is given by

$$E(x) = \mu = \sum_{x=0}^n f(x) = \sum_{x=0}^n x P(X=x)$$

$$= \sum_{x=0}^n x \cdot {}^n C_x p^x q^{n-x}$$

$$= \sum_{x=0}^n x \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

$$= \sum_{x=1}^n \frac{x n!}{(n-x)! (x-1)!} p^x q^{n-x}$$

$$= \sum_{x=1}^n \frac{n!}{(n-x)! (x-1)!} \cdot p^{(x-1)+1} q^{(n-1)-(x-1)}$$

$$= \sum_{x=1}^n \frac{n(n-1)!}{((n-1)-(x-1))!(x-1)!} p^{x-1} p q^{(n-1)-(x-1)}$$

$$= np \sum_{x=1}^n (n-1) \binom{n-1}{x-1} p^{x-1} q^{(n-1)-(x-1)} \quad \left\{ \text{Binomial form} \right\}$$

$$= np \sum_{x=1}^n (q+p)^{n-1} = np(1)^{n-1} = \underline{np}$$

$$\therefore \text{mean} = np$$

Variance of the binomial distribution

$$\text{W.K.T } \text{var}(X) = \sigma^2 = E(X^2) - (E(X))^2 = \sum_{x=0}^n x^2 f(x) - \mu^2$$

$$= \sum_{x=0}^n x^2 n C_x p^x q^{n-x} - (np)^2$$

$$= \sum_{x=0}^n (x^2 - x + x) n C_x p^x q^{n-x} - n^2 p^2$$

$$= \sum_{x=2}^n x(x-1) n C_x p^x q^{n-x} + \sum_{x=1}^n x n C_x p^x q^{n-x} - n^2 p^2$$

$$= \sum_{x=2}^n x(x-1) \frac{n!}{(n-x)!x!} p^x q^{n-x} + \sum_{x=0}^n x f(x) - n^2 p^2$$

$$= \sum_{x=2}^n \frac{x(x-1)}{(n-2)!(x-2)!} \frac{n!}{(n-x)!} p^x q^{n-x} + \mu - n^2 p^2$$

$$= \sum_{x=2}^n \frac{n!}{((n-2)-(x-2))!(x-2)!} \cdot p^{(x-2)+2} q^{(n-2)-(x-2)}$$

$$= \sum_{x=2}^n \frac{n(n-1)(n-2)!}{((n-2)-(x-2))!(x-2)!} \cdot p^{(x-2)} p^2 q^{(n-2)-(x-2)} + np - n^2 p^2$$

$$= n(n-1)p^2 \sum_{x=2}^n \frac{(n-2)!}{((n-2)-(x-2))!(x-2)!} \cdot p^{(x-2)} q^{(n-2)-(x-2)} + np - n^2 p^2$$

$$= n(n-1)p^2 \cdot (p+q)^{n-2} + np - n^2 p^2$$

$$= n^2 p^2 - np^2 + np - n^2 p^2 = np(1-p) = npq$$

$$\text{Variance} = npq$$

Mode of the Binomial Distribution

Mode of B.D. is given by integral part $((n+1)p)$ if $(n+1)p$ is not an integer.

If $(n+1)p$ is an integer, it will have 2 modes:

$$(n+1)p - 1, (n+1)p$$

In such a case, it is called Bimodal Distribution.

Q) The mean and variance of the binomial distribution is 3 and $9/4$. Find the value of $n, P(X \geq 7), P(1 \leq X \leq 6)$

$$\text{W.K.T mean} = np = 3$$

$$\text{Variance} = npq = \frac{9}{4} = 3(1-p) = \frac{9}{4}$$
$$\Rightarrow p = \frac{1}{4}$$

$$\therefore np = 3 \Rightarrow n\left(\frac{1}{4}\right) = 3 \Rightarrow n = 12$$

$$P(X \geq 7) = \{1 - P(X \leq 6)\} \Rightarrow P(X=7) + P(X=8) + \dots + P(X=12)$$

$$\begin{aligned} \Rightarrow P(X \geq 7) &= {}^{12}C_7 p^7 q^5 + {}^{12}C_8 p^8 q^4 + \dots \\ &= {}^{12}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^5 + {}^{12}C_8 \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^4 + {}^{12}C_9 \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right)^3 \\ &\quad + {}^{12}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^2 + {}^{12}C_{11} \left(\frac{1}{4}\right)^{11} \left(\frac{3}{4}\right)^1 + {}^{12}C_{12} \left(\frac{1}{4}\right)^{12} \left(\frac{3}{4}\right)^0 \\ &= \frac{1}{4^{12}} \left[3^5 {}^{12}C_7 + 3^4 {}^{12}C_8 + 3^3 {}^{12}C_9 + 3^2 {}^{12}C_{10} + 3^1 {}^{12}C_{11} + 3^0 {}^{12}C_{12} \right] \end{aligned}$$

$$P(X=x) = \begin{cases} {}^n C_x p^x q^{n-x}, & x = 0, 1, 2, 3, \dots, n \quad \text{non-negative finite values} \\ 0 & \text{otherwise} \end{cases}$$

$$B(x, n, p)$$

$$\mu = \text{mean} = np$$

$$\sigma^2 = \text{variance} = npq$$

$$\sigma = \text{S.D.} = \sqrt{npq}$$

19/9/25

Q. If 6 dice were thrown 729 times, how many times do you expect atleast 3 dice to show a five or six.

(i) If the prob. of a defective bolt is 0.2, find mean & S.D for the distribution of bolts in a total of 400.

(ii) If 10% of the rivets produced by the machine are defective, find the prob. that out of 5 rivets chosen at random,

- a) None will be defective
- b) One will be defective
- c) Atmost 2 rivets will be defective

(iii) Let 'X' represents defective rivets.

Probability of getting defective rivets is 10% = $0.1 = p$

$$\Rightarrow q = 1 - p = 0.9$$

Given $n = 5$

$$\text{i) } P(\text{none}) = P(X=0) = {}^nC_0 p^0 q^{n-0}$$

$$= {}^5C_0 (0.1)^0 (0.9)^5$$

$$= 9^5 \times 10^{-5} = 0.59049$$

$$\text{ii) } P(\text{one}) = P(X=1) = {}^nC_1 p^1 q^{n-1}$$

$$= {}^5C_1 (0.1)^1 (0.9)^4$$

$$= 5 \times 0.1 \times (0.9)^4$$

$$= 0.32805$$

$$\text{(iii) } P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= 0.59049 + 0.32805 + {}^5C_2 (0.1)^2 (0.9)^3$$

$$= 0.59049 + 0.32805 + 0.0729$$

$$= 0.99144$$

(ii) Let 'X' represent defective bolt.

$$\text{Given } p = 0.2$$

$$n = 400$$

$$q = 1 - p = 0.8$$

$$\text{Mean} = np = 400(0.2) \\ = 80$$

$$\text{S.D} = \sqrt{npq} = \sqrt{400(0.2)(0.8)} \\ = \sqrt{4 \times 16} \\ = 2 \times 4 \\ \text{S.D} = 8$$

(i) Let X represent getting 5 or 6 when a die is rolled.

$$P = P(5) + P(6)$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \rightarrow P = \frac{1}{3}, q = \frac{2}{3}$$

$$\text{Given } n = 6$$

Getting 5 or 6 on atleast 3 dice when thrown 1 time

$$\text{is } P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) + P(X=6)$$

$$= {}^nC_3 p^3 q^3 + {}^nC_4 p^4 q^2 + {}^nC_5 p^5 q + {}^nC_6 p^6 q^0$$

$$= {}^6C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 + {}^6C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 + {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^1 \\ + {}^6C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^0$$

$$= \frac{1}{3^6} \left[{}^6C_3 \cdot 2^3 + {}^6C_4 \cdot 2^2 + {}^6C_5 \cdot 2 + {}^6C_6 \right]$$

$$= \frac{1}{3^6} \left[\frac{6 \times 5 \times 4}{3!} \times 8 + \frac{6 \times 5 \times 4}{2!} \times 2 + 6 \times 2 + 1 \right]$$

$$= \frac{1}{3^6} \left[160 + 60 + 12 + 1 \right] = \frac{233}{3^6} = \frac{233}{729}$$

$$\begin{aligned} \text{for 729 times} &\Rightarrow 729 \times \frac{1}{3^6} [233] \\ &= 233 \\ &= \end{aligned}$$

Q) The prob of a man hitting a target is $\frac{1}{3}$. If he fire 5 times, i) what is the prob. of him hitting the target atleast twice.

ii) How many times must he fire so that the probability of him hitting the target atleast once is $> 90\%$.

Let 'X' represent hitting the target.

$$\text{Given that } p = \frac{1}{3}$$

$$\Rightarrow q = \frac{2}{3}$$

$$\Rightarrow \text{Given that } n = 5.$$

$$P(X \geq 2) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$\text{or } 1 - (P(X=0) + P(X=1))$$

$$= 1 - \left({}^5C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^5 + {}^5C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^4 \right)$$

$$= 1 - \left(\frac{25}{3^5} + \frac{5 \cdot 2^4}{3^5} \right)$$

$$= 1 - \left[\frac{32 + 80}{3^5} \right]$$

$$= 1 - \frac{112}{243}$$

$$= \frac{131}{243} = 0.53909$$

$$\text{(ii)} \quad n = ? \quad \text{such that } P(X \geq 1) > 90\%$$

$$\Rightarrow 1 - P(X < 1) > 90\%$$

$$= (1 - {}^nC_0 \cdot p^0 q^{n-0}) > 0.9$$

$$\cdot \left[1 - \left(\frac{1}{3} \right)^n \left(\frac{2}{3} \right)^n \right] > 0.9$$

$$\left(\frac{2}{3} \right)^n \leq 0.1$$

$$n \log \left(\frac{2}{3} \right) \leq \log(0.1)$$

$$n(-0.17609) \leq -1$$

$$n \geq 5.678$$

$$\boxed{n=6}$$

or) by trial and error

$$\left(\frac{2}{3} \right)^1 \leq 0.1 \quad \text{No}$$

$$0.66$$

$$\left(\frac{2}{3} \right)^2 = 0.444$$

$$\left(\frac{2}{3} \right)^3 = 0.296$$

$$\left(\frac{2}{3} \right)^4 = 0.1975$$

$$\left(\frac{2}{3} \right)^5 = 0.1316$$

$$\left(\frac{2}{3} \right)^6 = 0.08779 \leq 0.1$$

$$\therefore \boxed{n=6}$$

Q) It has been claimed that 60% of all solar heat installations, the utility bill is reduced by atleast $\frac{1}{3}$ rd. Accordingly, what are the probabilities the utility bill will be reduced by atleast $\frac{1}{3}$ rd in

(i) 4 of 5 installations

(ii) Atleast 4 of 5 installations.

Let 'X' represents reducing utility bill by $\frac{1}{3}$ rd.

$$\text{Given } p = 60\% = \frac{60}{100} = \frac{3}{5} \Rightarrow q = \frac{2}{5}$$

$$\text{Given } n = 5$$

$$P(X=x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$(i) P(X=4) = {}^5C_4 p^4 q^{5-4}$$

$$= {}^5C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)$$

$$= 5 \left[\frac{3^4 \cdot 2}{5^5} \right]$$

$$= \frac{162}{625} = 0.2592$$

$$(ii) P(X \geq 4) = P(X=4) + P(X=5)$$

$$= {}^5C_4 p^4 q^{5-4} + {}^5C_5 p^5 q^{5-5}$$

$$= 0.2592 + \frac{35}{5^5}$$

$$= 0.2592 + 0.07776$$

$$= 0.33696$$

Q12 die are thrown 120 times. Find the avg no of times in which no on the first die exceeds the number on the second die.

Given, $n = 120$.

No. on 1st die > No. on 2nd die

Total $\Rightarrow 36$

$$\text{Favorable} \Rightarrow \frac{5+4+3+2+1}{36} = \frac{15}{36} = p$$

$$\text{Mean} = np = \frac{120 \times 15}{36} = 50$$

Q) In 8 throws, of a die, 5 or 6 is considered a success. Find the mean no. of successes and the SD.

Given that $n = 8$

$$p = \frac{2}{6} = \frac{1}{3} ; q = \frac{2}{3}$$

$$\text{mean} = np = 8 \times \frac{1}{3} = \frac{8}{3}$$

$$\text{Variance} = npq = 8 \times \frac{1}{3} \times \frac{2}{3} = \frac{16}{9}$$

$$SD = \sqrt{\frac{16}{9}} = \frac{4}{3}$$

Q) A traffic control engineer reports that 75% of the vehicles passing through a checkpoint are from within a state. What is the prob. that fewer than 4 of the 9 are from out of the state.

Given that $n = 9$

$$p = 75\% = \frac{75}{100} = \frac{3}{4} \Rightarrow q = \frac{1}{4}$$

$$\Rightarrow p = \frac{3}{4}, q = \frac{1}{4}$$

$$P(X=4) = {}^9C_4 p^4 q^{9-4}$$

$$\Rightarrow {}^9C_4 \cdot \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^5$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} \cdot \frac{3^4}{4^4} \cdot \frac{1}{4}$$

$$= \frac{126 \times 243}{4^9} = 0.1168$$

Q) An irregular 6 faced die is thrown. And the expectation that in 10 throws, it gives 5 even numbers is twice the expectation that it will give 4 even numbers. How many times in 10000 sets of 10 throws would you expect to give no even number.

Given that $n=10$, $N=10,000$.

Let X represents getting even numbers.

$$P(X=5) = 2P(X=4)$$

$$= {}^{10}C_5 p^5 q^5 = 2 \cdot {}^{10}C_4 p^4 q^6$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6}{5! 5} = 2 \times \frac{10 \times 9 \times 8 \times 7 \times 6}{4! 4!} \cdot \frac{q}{p}$$

$$\frac{p}{q} = \frac{10}{6}$$

$$\frac{p}{q} = \frac{5}{3}$$

$$\frac{p}{1-p} = \frac{5}{3}$$

$$\Rightarrow 3p = 5 - 5p$$

$$8p = 5$$

$$p = 5/8 ; q = 3/8$$

$$P(X=0) \Rightarrow {}^{10}C_0 \cdot p^0 q^{10}$$

$$\Rightarrow {}^{10}C_0 \cdot \left(\frac{5}{8}\right)^0 \left(\frac{3}{8}\right)^{10} = 0.00054993$$

$$= \frac{3^{10}}{8^{10}} \times N$$

$$= \frac{3^{10}}{8^{10}} \times 10,000 = 0.54993$$

0.54993
 ≈ 1 time in 10000 times

Q) If the chance that any of the 10 telephone lines is busy at an instant is 0.2. What is the most probable number of busy lines? And what is the probability of that number?

Given that $p=0.2$, $q=0.8$, $n=10$

Most probable no. of busy line = avg no. of busy line

$$np = 10 \times 0.2 = \underline{\underline{2}}$$

FITTING BINOMIAL $\rightarrow$ NEXT CLASS

Q) Out of 800 families with 5 children each (could) how many would you expect to be

(i) 3 boys (ii) 5 girls (iii) Either 2 or 3 boys.

(iv) At least 1 boy, assume equal prob for B & G.

$$N = 800, n = 5.$$

$$p = \frac{1}{2}, q = \frac{1}{2}$$

Let X represent no. of boys.

$$(i) P(X=3) = {}^5C_3 \cdot \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{2^5} \cdot \frac{10}{32} = \frac{5}{16} \times 800 = 250$$

$$(ii) P(X=0) = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{2^5} = \frac{1}{32} \times 800 = 25$$

$$(iii) P(X=2) + P(X=3) = {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 + \frac{10}{2^5} \cdot \frac{5}{16} \\ + {}^5C_3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{10}{2^5} + \frac{5}{16} \\ = \frac{5}{8} \times 800 = 500$$

$$(iv) P(X \geq 1) \Rightarrow 1 - P(X < 1) = 1 - P(X=0) \\ = 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 \\ = 1 - \frac{1}{32} = \frac{31}{32} \times 800 = 775$$

Fitting Binomial Distribution

24/09/25

Recurrence Relation of Binomial Distribution:

$$P(x+1) = \frac{n-x}{x+1} \cdot \frac{p}{q} P(x)$$

To use this recurrence relation, we need to have first value i.e. $P(0)$, then we can find $P(1), P(2), \dots$

$P(0)$ can be found from $P(X=x) = {}^nC_x p^x q^{n-x}$

$$P(X=0) = {}^nC_0 p^0 q^{n-0}$$

$$P(0) = q^n$$

To get expected frequencies, multiply Recurrence relation with N (Total frequency).

$$N \cdot P(x+1) = \frac{n-x}{x+1} \cdot \frac{p}{q} N \cdot P(x)$$

$$\Rightarrow f(x+1) = \frac{n-x}{x+1} \cdot \frac{p}{q} \cdot f(x)$$

Q1 4 coins are tossed 160 times. The no. of times 2 heads appear is given below.

x	0	1	2	3	4
No. of times	8	34	69	43	6

$$N = \text{Total frequencies} = 8 + 34 + 69 + 43 + 6 = 160$$

$$\text{w.k.t } P(X=x) = {}^nC_x p^x q^{n-x}$$

$$P(X=0) = {}^nC_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{4-0} = \frac{1}{16}$$

$$f(0) = N P(X=0) = 160 \left(\frac{1}{16}\right) = 10$$

W.K.T R.R is $f(x+1) = \frac{n-x}{x+1} \cdot \frac{P}{Q} \cdot f(x)$

$$f(x+1) = \frac{4-x}{x+1} \cdot 1 \cdot f(x) \quad \text{--- (1)}$$

$$\left\{ \because p=q = \frac{1}{2} \Rightarrow \frac{P}{Q} = 1 \right\}$$

Put $x=0$ in (1)

$$f(0+1) = \frac{4-0}{0+1} f(0)$$

$$f(1) = 4f(0)$$

$$f(1) = 40$$

Put $x=1$ in (1)

$$f(1+1) = \frac{4-1}{1+1} \cdot f(1)$$

$$f(2) = \frac{3}{2} \cdot 40$$

$$f(2) = 60$$

Put $x=2$ in (1)

$$f(2+1) = \frac{4-2}{2+1} \cdot f(2)$$

$$= \frac{2}{3} \cdot 60$$

$$f(3) = 40$$

Put $x=3$

$$f(3+1) = \frac{4-3}{3+1} \cdot f(3)$$

$$= \frac{1}{4} \cdot 40$$

$$f(4) = 10$$

These are all expected frequencies. sum of all values should be 150. $f(0) + f(1) + f(2) + \dots + f(4) = 160$

Fit the binomial distribution to the following frequency distribution.

x	0	1	2	3	4	5
$f(x)$	2	14	20	34	22	8

$$n = 5$$

$$N = \sum f(x) = 2 + 14 + 20 + 34 + 22 + 8 \\ = 100$$

$$\text{Mean} = \mu = \frac{\sum x f(x)}{\sum f(x)}$$

$$\mu = np = \frac{0 \times 2 + 1 \times 14 + 2 \times 20 + 3 \times 34 + 4 \times 22 + 5 \times 8}{100}$$

$$np = 2.84$$

$$p = \frac{2.84}{5} = 0.568$$

$$q = 1 - p = 1 - 0.568 = 0.432$$

$$P(x=0) = {}^n C_0 p^0 q^{n-0} = {}^5 C_0 p^0 q^5 \\ = (0.432)^5 \\ = 0.0150$$

$$f(0) = N \times P(0) = 100 \times 0.0150$$

$$f(0) = 1.5 \approx 2$$

$$\frac{p}{q} = \frac{0.568}{0.432} = 1.3148$$

$$f(x+1) = \frac{5-x}{x+1} \cdot (1.3148) \cdot f(x) \quad \text{--- (1)}$$

Put $x=0$ in (1)

$$f(1) = \frac{5}{1} (1.3148)(1.5)$$

$$= 9.861 \approx 10$$

$$f(2) = \frac{5-1}{1+1} (1.3148)(9.861)$$

$$= 25.9304 \approx 26$$

$$f(3) = \frac{5-2}{2+1} (1.3148)(25.9304) = 34.0932 \approx 34$$

$$f(4) = \frac{5-3}{3+1} (1.3148)(34.0932) = 22.4128 \approx 22$$

$$f(5) = \frac{5-4}{4+1} (1.3148)(22.4128) = 5.8936 \approx 5$$

$$\rightarrow 2+10+26+34+22+5=100$$

(a)

Expected frequencies are given by

$$N (q+p)^n$$

$$N=100$$

$$p=0.568, q=0.432$$

$$n=5$$

$$= 100 \times (q+p)^5$$

$$= 100 ({}^5C_0 p^0 q^5 + {}^5C_1 p^1 q^4 + {}^5C_2 p^2 q^3 + {}^5C_3 p^3 q^2 + {}^5C_4 p^4 q^1 + {}^5C_5 p^5 q^0)$$

$$= 100 (1 \times (0.568)^0 (0.432)^5 + 5 (0.568) (0.432)^4$$

$$+ 10 (0.568)^2 (0.432)^3 + 10 (0.568)^3 (0.432)^2$$

$$+ 5 (0.568)^4 (0.432)^1 + (0.568)^5)$$

$$= 100 (0.0150 + 0.0989 + 0.2601 + 0.3419 +$$

$$0.2248 + 0.0591)$$

$$= 1.5 + 9.89 + 26.01 + 34.19 + 22.48 + 5.91$$

$$\approx 100$$

Poisson's Distribution Does not apply until Binomial fails

When no. of trials 'n' is large and 'p' is small, then it is difficult to use Binomial Distribution.

In such a case, we use poisson's approach to Binomial Distribution.

Here, we assume as $n \rightarrow \infty$, $p \rightarrow 0$ so that the product $np = \text{finite} = \lambda$ (say)

→ The variable 'x' is said to follow poisson distribution if it takes non-negative values and its probability mass function is given by

$$P(X=x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & , x=0,1,2,\dots \\ 0 & \text{otherwise (for -ve values)} \end{cases}$$

Here, λ is the parameter of poisson distribution.

(Only use poisson when $p < n$)

ST mean and variance of poisson distribution are equal.

Since poisson is a discrete distribution, its mean is

$$\mu = \sum_{x=0}^{\infty} x P(X=x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-1)!} = e^{-\lambda} \cdot \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$\text{w.k.t } e^x = 1 + x + \frac{x^2}{2!} + \dots = e^{-\lambda} \lambda e^{\lambda}$$

$$\boxed{\text{Mean} = \mu = \lambda}$$

$$\text{Variance} = \sum_{x=0}^{\infty} x^2 f(x) - \mu^2$$

$$\sigma^2 = \sum_{x=0}^{\infty} (x^2 - x + x) f(x) - \mu^2$$

$$= \sum_{x=0}^{\infty} x(x-1) f(x) + \sum_{x=0}^{\infty} x f(x) - \mu^2$$

$$= \sum_{x=2}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} + \mu - \mu^2$$

$$= e^{-\lambda} \lambda^2 \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda - \lambda^2$$

$$e^{-\lambda} \lambda^2 \cdot e^{\lambda} + \lambda - \lambda^2 = \lambda$$

$$\text{Variance} = \sigma^2 = \lambda$$

$$\therefore \text{Mean} = \text{Variance} = \lambda$$

$$\text{S.D} = \sqrt{\lambda}$$

Q. If the prob that an individual suffers a bad reaction from a certain infect is 0.001. Determine the prob that out of 2000 individuals $\{n \gg p\}$

- (i) Exactly 3 (ii) More than 2 individuals (iii) none
(iv) More than one individual suffer a bad reaction.

$$\lambda = np = 2000 \times 10^{-3} = 2$$

$$(i) P(X=3) = \frac{e^{-2} 2^3}{3!}$$

$$(ii) P(X > 2) = 1 - P(X \leq 2)$$

$$(or) 1 - \left(\frac{e^{-2} 2^0}{0!} + \frac{e^{-2} 2^1}{1!} + \frac{e^{-2} 2^2}{2!} \right)$$

$$= 1 - e^{-2} - 2e^{-2} - 2e^{-2}$$

$$= 1 - 5e^{-2}$$

$$(iii) P(X=0) = \frac{e^{-2} 2^0}{0!} = e^{-2}$$

$$(iv) P(X \geq 2) = 1 - P(X < 2) \text{ or } P(X > 1) = 1 - P(X \leq 1)$$

$$= 1 - (e^{-2} + 2e^{-2}) = 1 - 3e^{-2}$$

Q3 A car hire firm has 2 cars which it hires out day by day. The no. of demands for a car on each day is distributed as a poisson distribution with mean 1.5. Calculate the proportion of days

i) on which there is no demand

ii) on which demand is refused

X = demand for car

$$i) P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda} = e^{-1.5}$$

$$\lambda = 1.5$$

ii) $P(\text{refused}) = P(\text{more than 2 cars is asked, then company refuses})$

$$= P(X > 2)$$

$$= 1 - P(X \leq 2)$$

$$= 1 - P(X=0) - P(X=1) - P(X=2)$$

$$= 1 - e^{-1.5} - \frac{e^{-1.5} (1.5)^1}{1!} - \frac{e^{-1.5} (1.5)^2}{2!}$$

$$= 1 - e^{-1.5} - (1.5)e^{-1.5} - (1.5)^2 e^{-1.5}$$

$$= 1 - e^{-1.5} (1 + 1.5 + (1.5)^2)$$

2

Q4 If X is a poisson variate such that

$$3P(X=4) = \frac{1}{2} P(X=2) + P(X=0) \text{ Find mean, } P(X \leq 2)$$

$$\frac{3e^{-\lambda} \lambda^4}{4!} = \frac{1}{2} \frac{e^{-\lambda} \lambda^2}{2!} + \frac{e^{-\lambda} \lambda^0}{0!}$$

$$\frac{3\lambda^4}{24} = \frac{\lambda^2}{4} + 1$$

$$3\lambda^4 - 6\lambda^2 - 24 = 0$$

$$\lambda^2 = \frac{6 \pm \sqrt{36 + 288}}{6}$$

$$= \frac{6 \pm 18}{6} = \frac{24}{6}, \left\{ \frac{12}{6} \right\} \text{ or } \frac{24}{6}$$

$$\therefore \lambda^2 = 4$$

$$\lambda = \pm 2$$

$$\text{mean} = \lambda = 2$$

$$P(X \leq 2) = \frac{e^{-2} \cdot 1}{1} + \frac{e^{-2} (2)^1}{1!} + \frac{e^{-2} 2^2}{2!}$$

$$= e^{-2} + 2e^{-2} + 2e^{-2}$$

$$= 5e^{-2}$$

08/10/25

Q3 2% of the items in a factory are defective. The items are packed in boxes. What is the probability that there are (i) Two defective items (ii) At least 3 defective items out of 100 items.

Let X represent defective items.

$$p = 2\% = 0.02$$

$$n = 100$$

$$\Rightarrow np = 100 \times 2\% = 2, \quad P(X=x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x=0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

$$\lambda = np = 2$$

$$(i) 2 \text{ defective} = P(X=2) = \frac{e^{-\lambda} \lambda^2}{2!} = \frac{e^{-2} 2^2}{2!} = 2e^{-2}$$

$$(ii) \text{At least 3 defective} = P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - P(X=0) - P(X=1) - P(X=2)$$

$$= 1 - \frac{e^{-2} 2^0}{0!} - \frac{e^{-2} 2^1}{1!} - \frac{e^{-2} 2^2}{2!} = 1 - e^{-2} [1 + 2 + 2]$$

Recurrence Relation (R.R) of Poisson Distribution:

$$p(x+1) = \frac{\lambda}{x+1} p(x)$$

It can be provided $p(0)$ is known, used,

$$p(x+1) = \frac{\lambda}{x+1} p(x)$$

Q] Fit the Poisson Distribution for the following data and calculate the expected frequencies

x	0	1	2	3	4
$f(x)$	109	65	22	3	1

$$\text{W.K.T. mean} = \lambda = \frac{\sum x_i f(x_i)}{\sum f(x_i)} \quad \text{or} \quad \frac{\sum x f(x)}{\sum f(x)}$$

↳ N total freq.

$$= \frac{0 \times 109 + 1 \times 65 + 2 \times 22 + 3 \times 3 + 4 \times 1}{109 + 65 + 22 + 3 + 1}$$

$$= \frac{0 + 65 + 44 + 9 + 4}{200} = \frac{122}{200}$$

$$= 0.61$$

$$\text{W.K.T. } P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$p(x=0) = p(0) = \frac{e^{-0.61} \cdot \lambda^0}{0!} = e^{-0.61} = 0.5433$$

$$f(0) = N \cdot p(0)$$

$$= 200(0.5433)$$

$$= 108.66 \approx 109$$

RR of PD is

$$f(x+1) = \frac{\lambda}{x+1} f(x)$$

$$f(x+1) = \frac{0.61}{x+1} f(x) \quad \text{--- (1)}$$

Put $x=0$ in (1)

$$\begin{aligned} f(1) &= 0.61 f(0) \\ &= 0.61 \times 108.66 \end{aligned}$$

$$f(1) = 66.2826 \approx 66$$

Put $x=1$ in (1)

$$\begin{aligned} f(2) &= \frac{0.61}{2} f(1) \\ &= \frac{0.61 \times 66.2826}{2} \end{aligned}$$

$$f(2) = 20.216 \approx 20$$

Put $x=2$ in eq (1)

$$\begin{aligned} f(3) &= \frac{0.61}{3} \times f(2) \\ &= \frac{0.61 \times 20.216}{3} \\ &= 4.110 \approx 4 \end{aligned}$$

Put $x=3$ in eq (1)

$$\begin{aligned} f(4) &= \frac{0.61}{4} f(3) \\ &= \frac{0.61}{4} \times 4.110 \\ &= 0.626 \approx 1 \end{aligned}$$

check: $109 + 66 + 20 + 4 + 1 = 200$ ✓ $\therefore$ Right ✓

The expected frequencies are: 109, 66, 20, 4, 1

x	0	1	2	3	4
obs $f(x)$	109	65	22	3	1
exp $f(x)$	109	66	20	4	1